

Use the following context to answer questions 1-5.

The length of a machine part should be 6.75 inches (in). The manufacturing tolerance is within 0.04 in.

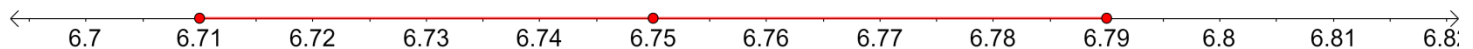
1. What is the minimum value the machine part could be?

$$6.75 - 0.04 = 6.71 \text{ in.}$$

2. What is the maximum value the machine part could be?

$$6.75 + 0.04 = 6.79 \text{ in.}$$

3. Sketch a number line that shows the minimum value, maximum value, and 6.75.



4. Write an absolute value equation that could be used to determine the maximum and minimum value of m , the length of the machine part, algebraically.

$$|m - 6.75| = 0.04$$

5. Show that the solutions to the equation match the maximum and minimum values determined in questions 1 and 2?

$$m - 6.75 = 0.04$$

$$+6.75 \quad +6.75$$

$$m = 6.79$$

$$m - 6.75 = -0.04$$

$$+6.75 \quad +6.75$$

$$m = 6.71$$

and

For questions 6-10, solve each absolute value equation.

6. $0 = 4|x + 3| - 16$

$$\begin{aligned} &+16 \quad +16 \\ \frac{16}{4} &= \frac{4|x+3|}{4} \\ 4 &= |x+3| \\ x+3 &= 4 \quad \text{and} \quad x+3 = -4 \\ -3 \quad -3 & \quad -3 \quad -3 \\ x &= 1 \quad \quad \quad x = -7 \\ &\quad \quad \quad \boxed{x = 1 \text{ and } -7} \end{aligned}$$

7. $|-2x + 7| - 21 = 0$

$$\begin{aligned} &+21 \quad +21 \\ |-2x+7| &= 21 \\ -2x+7 &= 21 \quad \text{and} \quad -2x+7 = -21 \\ -7 \quad -7 & \quad -7 \quad -7 \\ \frac{-2x}{-2} &= \frac{14}{-2} \quad \quad \quad \frac{-2x}{-2} = \frac{-28}{-2} \\ x &= -7 \quad \quad \quad x = 14 \\ &\quad \quad \quad \boxed{-7 \text{ and } 14} \end{aligned}$$

8. $|x + 3| + 17 = 11$

$$\begin{aligned} &-17 \quad -17 \\ |x+3| &= -6 \\ \boxed{\text{No solution}} \end{aligned}$$

9. $|4x + 5| - 8 = 29$

$$\begin{aligned} &+8 \quad +8 \\ |4x+5| &= 37 \\ 4x+5 &= 37 \quad \text{and} \quad 4x+5 = -37 \\ -5 \quad -5 & \quad -5 \quad -5 \\ \frac{4x}{4} &= \frac{32}{4} \quad \quad \quad \frac{4x}{4} = \frac{-42}{4} \\ x &= 8 \quad \quad \quad x = -10.5 \\ &\quad \quad \quad \boxed{-10.5 \text{ and } 8} \end{aligned}$$

10. $-|x - 2| + 9 = 0$

$$\begin{aligned} &-9 \quad -9 \\ \frac{-|x-2|}{-1} &= \frac{-9}{-1} \\ |x-2| &= 9 \\ x-2 &= 9 \quad \quad x-2 = -9 \\ +2 \quad +2 & \quad +2 \quad +2 \\ x &= 11 \quad \quad \quad x = -7 \\ &\quad \quad \quad \boxed{-7 \text{ and } 11} \end{aligned}$$